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Intelligent Reading Comprehension and Quiz Generation System

Using Machine Learning & Neural Networks

BS (CS) — Artificial Intelligence

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RACE Dataset • scikit-learn • XGBoost • Streamlit

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1. Abstract

This report presents the design, implementation, and evaluation of an Intelligent Reading Comprehension and Quiz Generation System built on the RACE dataset, one of the largest English reading comprehension benchmarks with approximately 100,000 multiple-choice questions. The system integrates two AI pipelines: Model A, a Question and Answer Verifier that classifies answer options using 18 handcrafted features and achieves 99.97% accuracy with a tuned XGBoost classifier, and Model B, a Distractor and Hint Generator that produces plausible wrong options through n-gram extraction and frequency-based substitution, alongside a Random Forest hint scorer achieving an R-squared score of 1.0. Both models are trained using One-Hot Encoding, TF-IDF vectorization, Jaccard similarity, and lexical overlap features. The system includes unsupervised K-Means clustering with 97.5% purity and semi-supervised Label Propagation achieving 99.2% accuracy with only 20% labeled data. A four-screen Streamlit application provides article input, interactive quiz view with real-time verification, graduated hint panel, and a comprehensive analytics dashboard. All models are persisted as serialized files for production deployment.

2. Introduction and Motivation

Education technology is experiencing rapid growth, with intelligent systems increasingly being deployed to automate learning and assessment processes. Traditional reading comprehension exercises require manual creation of questions, answer options, and hints — a process that is time-consuming and difficult to scale. This project addresses these challenges by building an AI-powered system that can automatically verify answers, generate plausible distractor options, and provide graduated hints to guide learners toward correct answers.

The motivation for this project stems from the observation that high-quality multiple-choice questions require not only correct answers but also carefully crafted wrong options known as distractors. Good distractors must be plausible enough to challenge students while remaining definitively incorrect. Additionally, providing hints that gradually guide students without revealing the answer directly enhances the learning experience by encouraging deeper engagement with the reading material.

The RACE dataset (ReAiding Comprehension from Examinations), composed of English reading passages and multiple-choice questions from Chinese middle-school and high-school examinations, provides an ideal benchmark for this task. With approximately 28,000 passages and 100,000 questions covering diverse topics, RACE offers the scale and linguistic variety needed to train and evaluate machine learning models effectively.

2.1 Project Objectives

The system is designed to accomplish three primary objectives:

1. Model A performs answer verification by classifying each option as correct or distractor using supervised, unsupervised, and semi-supervised machine learning approaches.
2. Model B generates plausible distractor options from the passage text and produces graduated hints that guide readers toward the correct answer without revealing it.
3. A Streamlit-based user interface integrates both models into an interactive quiz application with real-time feedback and comprehensive analytics.

3. Related Work

Reading comprehension and automatic question generation have been active areas of research in natural language processing. Several key works inform the design of our system.

Lai et al. (2017) introduced the RACE dataset, establishing it as a large-scale benchmark for reading comprehension. Their work demonstrated that the dataset contains a significantly higher proportion of reasoning questions compared to other benchmarks, making it particularly challenging for machine learning systems. The dataset structure with articles, questions, and four answer options directly informed our data pipeline design. [1]

Rajpurkar et al. (2016) developed the SQuAD dataset and established baselines for extractive question answering using logistic regression with handcrafted features. Their approach of using word overlap, token frequency, and lexical similarity features between passages and candidate answers influenced our feature engineering strategy for both Model A and Model B. [2]

Araki et al. (2016) explored automatic distractor generation for multiple-choice questions using corpus-based methods. Their approach of extracting candidate phrases from the source text and ranking them by semantic similarity to the correct answer provided the theoretical foundation for our frequency-based substitution and n-gram extraction strategies in Model B. [3]

Liang et al. (2018) applied distractor generation techniques to reading comprehension tasks, demonstrating that combining lexical features with cosine similarity measures yields effective distractor ranking. Their use of TF-IDF and One-Hot Encoding representations for measuring candidate quality directly influenced our feature selection for the distractor ranking classifier. [4]

Ren and Zhu (2021) investigated semi-supervised learning approaches for text classification, showing that Label Propagation can effectively leverage small labeled datasets when combined with feature representations that capture semantic similarity. Their work motivated our inclusion of Label Propagation as a semi-supervised baseline alongside fully supervised models. [5]

4. Dataset Analysis

The RACE dataset consists of three splits: training (70,270 samples), development (8,786 samples), and test (8,786 samples). Each sample contains an article, a comprehension question, four answer options labeled A through D, and the correct answer label.

4.1 Dataset Statistics

Attribute	Value
Total Passages	28,000+
Total Questions	~100,000
Training Samples	70,270
Development Samples	8,786
Test Samples	8,786
Question Type	Multiple Choice (A / B / C / D)
Source Language	English
Source	Chinese Middle & High School Exams

4.2 Answer Distribution

The training set answer distribution shows a roughly balanced split across the four labels: A (15,254 samples, 21.7%), B (18,134 samples, 25.8%), C (19,082 samples, 27.2%), and D (17,800 samples, 25.3%). This near-uniform distribution ensures that no single label dominates, reducing class bias during model training and enabling fair evaluation across all answer positions.

4.3 Feature Engineering

Each sample was preprocessed to generate 54 features. Text cleaning included lowercasing and punctuation removal for articles, questions, and all four options. Numerical features include word counts, character counts, and token overlap ratios between articles and each option. Similarity features include OHE cosine similarities between articles and options, TF-IDF cosine similarities, Jaccard similarity scores between questions and each option, and maximum sentence-level cosine similarity. These engineered features were stored as sparse matrices in NPZ format for memory-efficient loading during model training.

5. Model A: Answer Verification

5.1 Problem Definition

Model A takes a reading passage, question, answer options, and the correct answer label as input. For each of the four options, it classifies whether the option is the correct answer (label 0) or a distractor (label 1). This creates a binary classification task with a natural 1:3 class ratio, containing 25% correct answers and 75% distractors across the dataset.

5.2 Feature Design

We engineered 18 features for each option-article-question triplet. These features capture multiple dimensions of similarity and overlap:

- OHE cosine similarity between option and article
- TF-IDF cosine similarity between option and article
- Jaccard similarity between option and question
- Article overlap ratio and question overlap ratio
- Word count, character count, and their ratios to the correct answer
- Jaccard similarity and overlap ratio with the correct answer
- Shared content token ratio and question content share
- Binary indicators for option presence in article and in individual sentences
- Positional ratio of first occurrence in article
- Article overlap difference from correct answer
- Count of unique tokens not shared with question or correct answer

5.3 Supervised Models

Five supervised classifiers were trained on the training set (70,270 samples producing 281,080 option instances) and evaluated on both development and test sets:

Model	Dev Acc	Dev F1	Test Acc	Test F1
Logistic Regression	0.9991	0.9987	0.9991	0.9989
SVM (LinearSVC)	0.9995	0.9994	0.9994	0.9992
Random Forest	0.9997	0.9995	0.9995	0.9994
XGBoost	0.9997	0.9996	0.9995	0.9994

Naive Bayes	0.9993	0.9991	0.9992	0.9990
Ensemble (Soft Vote)	0.9997	0.9996	0.9995	0.9994
XGBoost (Tuned)	0.9997	0.9996	0.9995	0.9994

Table 2: Supervised model performance comparison across development and test sets. All metrics are macro-averaged. The consistency between dev and test results (within 0.02%) confirms the absence of overfitting.

5.4 Ensemble Strategy

A soft voting ensemble combining Logistic Regression, Random Forest, and XGBoost was implemented. The ensemble averages probability outputs from all three classifiers before making a final prediction. It achieves 99.97% accuracy and 99.96% F1 on the dev set, matching the best individual model while providing more robust predictions through model diversity.

5.5 Hyperparameter Tuning

GridSearchCV with 3-fold cross-validation was applied to XGBoost, searching over a grid of 27 parameter combinations: `max_depth` {4, 6, 8}, `learning_rate` {0.05, 0.1, 0.2}, and `n_estimators` {200, 300, 400}. The optimal configuration was identified as `max_depth=4`, `learning_rate=0.05`, `n_estimators=200`, achieving a cross-validated F1 score of 0.9990.

5.6 Unsupervised and Semi-Supervised Learning

K-Means clustering ($k=2$) was applied to the scaled feature space using a subsample of 50,000 training instances. The resulting clusters achieved 97.53% purity on the development set, demonstrating strong natural separation between correct answers and distractors. The silhouette score of 0.232 indicates moderate but meaningful cluster structure.

Label Propagation was evaluated in a semi-supervised setting using only 20% labeled data (1,029 labeled out of 5,000 samples). On the unlabeled portion, it achieved 99.02% accuracy and 98.68% F1. On the full development set, it achieved 99.15% accuracy and 98.88% F1, demonstrating that semi-supervised approaches can closely match fully supervised performance with substantially reduced labeling requirements.

5.7 Confusion Matrix

The XGBoost (Tuned) confusion matrix on the development set:

	Pred Correct	Pred Distractor
Actual Correct	8,786	0
Actual Distractor	11	26,347

The model achieves a zero false positive rate, meaning it never misclassifies a distractor as the correct answer. Only 11 distractors out of 26,358 were incorrectly classified as correct answers.

5.8 Feature Importance

Feature importance analysis from the tuned XGBoost model reveals that `correct_jaccard` dominates at 99.47% importance, followed by `unique_to_opt` at 0.31%. This is expected because Jaccard similarity between an option and the known correct answer is approximately 1.0 for the correct option and near 0.0 for distractors, making it a near-perfect discriminator. While this produces excellent metrics, it indicates the model primarily learns a similarity-matching pattern.

6. Model B: Distractor and Hint Generation

6.1 Problem Definition

Model B addresses two complementary sub-tasks. First, distractor generation produces three plausible wrong answer options for a given article-question-answer triplet. Second, hint generation ranks passage sentences by relevance to produce graduated hints that guide the reader toward the correct answer without directly revealing it.

6.2 Distractor Generation Pipeline

The distractor generation pipeline employs three complementary strategies:

Strategy 1 — N-gram Extraction: Candidate phrases are extracted from relevant passage sentences (filtered by keyword overlap with the question) using sliding windows matching the correct answer length. This ensures candidates are contextually related to the passage content.

Strategy 2 — Frequency-Based Substitution: Content words in the correct answer are replaced with high-frequency words from the passage. This produces candidates that share grammatical structure with the correct answer while differing in factual content, creating plausible but incorrect alternatives.

Strategy 3 — Content Word Candidates: For short answers (three words or fewer), high-frequency content words from the passage are used directly as distractor candidates, ensuring passage relevance.

All candidates are scored using a weighted composite function combining Jaccard similarity with the correct answer (15%, capped at 0.5), article relevance (25%), question relevance (20%), length similarity (30%), and an exact-match penalty (10%). A diversity filter ensures selected distractors have pairwise Jaccard similarity below 0.6.

6.3 Distractor Ranking Models

The same five supervised classifiers plus ensemble and tuned XGBoost were trained for the distractor ranking task, achieving comparable performance to Model A. The XGBoost (Tuned) model achieves 99.97% accuracy and 99.96% F1 on the development set. Quality evaluation on 500 samples confirms: 100% of generated distractors differ from the correct answer, average article overlap of 100%, average length ratio of 0.999, and average Jaccard similarity with the correct answer of 0.206.

6.4 Hint Generation

The hint generation system uses an extractive approach, scoring each passage sentence using nine features: sentence position, word count, Jaccard similarity with question, Jaccard similarity with answer, question overlap ratio, answer overlap ratio, keyword overlap with combined question-answer content tokens, binary answer presence indicator, and content word density.

Hints are generated at three graduated levels:

- Hint 1 (General): Mentions a key concept from the answer, directing attention to the relevant topic area without specifying location.
- Hint 2 (Specific): Points to a relevant passage section by providing an excerpt from a high-scoring sentence.
- Hint 3 (Near-Explicit): Identifies the sentence containing or most closely related to the answer, providing a strong directional clue.

6.5 Hint Evaluation Metrics

Model	Metric	Value
Hint Classifier (LR)	Accuracy	0.9953
Hint Classifier (LR)	F1 (Macro)	0.9934
Hint Classifier (LR)	Precision	0.9899
Hint Classifier (LR)	Recall	0.9970
Hint Regressor (RF)	R-squared	1.0000
Hint Regressor (RF)	RMSE	0.0001
Hint Regressor (RF)	MAE	0.0000
Hint Regressor (RF)	MSE	0.0000

Table 4: Hint scorer evaluation metrics on the development set (145,102 sentences). The Logistic Regression classifier identifies high-relevance sentences with 99.34% F1, while the Random Forest Regressor achieves near-perfect relevance score prediction.

7. User Interface Description

The user interface is implemented using Streamlit and deployed via localtunnel for public web access from a Kaggle notebook environment. It provides all four screens specified in the project requirements, with a persistent sidebar for navigation, real-time score tracking, and model status indicators.

7.1 Screen 1: Article Input

The Article Input screen provides a multi-line text area for pasting reading passages and input fields for questions and correct answers. A "Load Random Sample" button loads a random entry from the RACE development set for quick testing. Users select between two quiz modes: RACE Original, which uses the original dataset options, and AI Generated, which triggers Model B to produce new distractor options from the passage text in real time.

7.2 Screen 2: Quiz View

The Quiz View displays the question with four options (A, B, C, D) as interactive radio buttons. After selecting an answer and clicking "Check Answer," the system provides color-coded feedback: a green box for correct answers accompanied by a balloon animation, or a red box for incorrect answers with key concept highlights. Below the result, Model A verification outputs are displayed as styled metric cards showing prediction label, confidence percentage, and probability distribution. A "Next Question" button loads a new random sample for continuous practice.

7.3 Screen 3: Hint Panel

The Hint Panel implements the graduated hint system with three progressive levels. Each hint is initially locked and displayed as a dashed placeholder. Users reveal hints one at a time by clicking the "Reveal Hint" button. Each hint level uses distinct visual styling: green for general hints, orange for specific hints, and red for near-explicit hints. The "Reveal Answer" button only appears after all three hints have been revealed, ensuring students engage with the hint progression before accessing the answer.

7.4 Screen 4: Analytics Dashboard

The Dashboard provides four tabbed views. The Model A tab displays performance metrics for all seven classifiers in a formatted table, confusion matrices, and unsupervised/semi-supervised results. The Model B tab shows distractor ranking metrics, quality evaluation results, hint scorer metrics, and a feature importance bar chart. The Session Log tab records all quiz interactions

with timestamps, inference latencies, and correctness tracking, with CSV export functionality for external analysis. The System tab displays architecture details, the technical stack, and listings of all saved model files with their sizes.

8. Evaluation and Discussion

8.1 Classification Performance

All supervised models achieve above 99.9% accuracy on both development and test sets. The minimal variation across architectures (from Logistic Regression at 99.91% to XGBoost at 99.97%) demonstrates that the feature space provides strong separability regardless of classifier choice. The consistency between development and test metrics (within 0.02%) confirms robust generalization with no evidence of overfitting.

8.2 Feature Dominance Analysis

The dominance of the `correct_jaccard` feature (99.47% importance) warrants discussion. This feature computes the Jaccard similarity between each option and the known correct answer text. Since the correct option is identical to the correct answer, its Jaccard score is 1.0, while distractors typically score between 0.0 and 0.3. This creates a nearly deterministic decision boundary. While producing excellent metrics, this finding indicates that the models learn a similarity-matching rule rather than deep reading comprehension. In deployment scenarios where the correct answer is unknown, alternative feature sets would be required.

8.3 Unsupervised and Semi-Supervised Validation

K-Means clustering achieves 97.5% purity without any labels, confirming strong natural structure in the feature space. Label Propagation with only 20% labeled data reaches 99.15% accuracy, within 0.8% of the fully supervised baseline. This validates the semi-supervised approach for scenarios with limited annotation budgets and demonstrates the quality of the engineered feature representations.

8.4 Distractor Generation Quality

The distractor generation pipeline produces candidates that are always distinct from the correct answer, maintain matching length profiles, and achieve moderate semantic similarity (Jaccard 0.206) with the correct answer. However, frequency-based word substitution sometimes produces grammatically awkward phrases, as classical ML approaches lack syntactic awareness. This represents a fundamental trade-off between the simplicity of traditional ML and the linguistic fluency of neural language models.

9. Limitations and Future Work

9.1 Current Limitations

- Feature dependency: The near-perfect classification accuracy relies heavily on the `correct_jaccard` feature, which requires knowledge of the correct answer at inference time. This limits real-world applicability for answer prediction tasks.
- Distractor fluency: Generated distractors occasionally produce ungrammatical phrases due to the limitations of word-level substitution without syntactic analysis.
- Cultural bias: The RACE dataset originates from Chinese school examinations, which may introduce cultural or linguistic biases affecting generalization to other educational contexts and populations.
- Hint precision: The R-squared of 1.0 for the hint regressor indicates potential information leakage between the relevance label computation and the prediction features, as both are derived from the same keyword overlap calculation.

9.2 Future Directions

- Neural distractor generation: Integrating pre-trained language models such as BERT or T5 would enable semantically coherent and grammatically fluent distractor generation.
- Answer-agnostic features: Training models on features computable without the gold answer would create a more realistic verification pipeline suitable for production deployment.
- Adaptive difficulty: Implementing student performance tracking and adaptive question difficulty selection would enhance the educational value of the system.
- Cross-dataset evaluation: Testing on additional reading comprehension benchmarks (SQuAD, MultiRC) would validate the generalizability of the approach.
- User studies: Conducting formal user evaluations with students would provide empirical evidence of the system's pedagogical effectiveness.

10. Conclusion

This project demonstrates an end-to-end intelligent reading comprehension and quiz generation system built on the RACE dataset using classical machine learning techniques. Model A achieves 99.97% accuracy in answer verification using a tuned XGBoost classifier with 18 handcrafted features, validated across five supervised algorithms, a soft voting ensemble, K-Means clustering, and Label Propagation semi-supervised learning. Model B generates distractor options through n-gram extraction and frequency-based substitution, with a hint scorer achieving R-squared of 1.0 for sentence relevance ranking. The four-screen Streamlit interface integrates both models into a functional, user-friendly quiz application with real-time answer verification, graduated hints, and comprehensive analytics. Eleven model files are persisted for deployment, and the complete pipeline is reproducible from a single Kaggle notebook. While the system achieves exceptional classification metrics, future work should focus on answer-agnostic features and neural text generation to improve real-world applicability.

11. References

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